

Blockchain-Based Cooperative System: A Case Study of Decentralized Autonomous Organization Implementation on the DChain Network

^{1st}Yitzhak Edmund Tio Manalu

Department of Electrical and Information Engineering
Universitas Gadjah Mada
Yogyakarta, Indonesia
yitzhakedmundtiomanalu@mail.ugm.ac.id

^{2nd}Ir. Noor Akhmad Setiawan, S.T., M.T., Ph.D., IPM.

Department of Electrical and Information Engineering
Universitas Gadjah Mada
Yogyakarta, Indonesia
noorwewe@ugm.ac.id

Abstract—Indonesian cooperatives (*koperasi*) face persistent operational challenges, opaque governance, manual treasury management, bureaucratic loan processing, and inconsistent surplus distribution, rooted in the absence of enforceable on-chain rules. Blockchain-based Decentralized Autonomous Organizations (DAOs) offer programmatic governance and transparency, yet existing implementations adopt token-weighted voting that contradicts the cooperative principle of one-member-one-vote mandated by Undang-Undang No. 25 Tahun 1992. This paper presents Lakomi, a blockchain-based cooperative system that encodes 26 provisions of UU 25/1992 into four EVM-compatible smart contracts deployed on DChain, an Indonesian academic consortium blockchain. Employing a Design Science Research methodology, the system implements a one-member-one-vote governance model with 67% quorum, six-category SHU distribution, role-based elections, and pengawas audit functions. Functional verification demonstrates 100% compliance across all 26 legal provisions. Gas profiling shows deployment costs of 1.58M–2.22M gas per contract, with key operations executing within 50K–250K gas. A feature comparison with traditional *koperasi* and token-based DAOs confirms that Lakomi uniquely satisfies both cooperative legal requirements and blockchain transparency guarantees.

Index Terms—blockchain, cooperative, *koperasi*, DAO, smart contract, UU 25/1992, one-member-one-vote, SHU, DChain, compliance

I. INTRODUCTION

Indonesia’s cooperative (*koperasi*) movement, rooted in *gotong royong* (mutual cooperation), encompasses over 127,000 registered cooperatives serving 27 million members nationwide. These cooperatives manage member savings (*simpanan*), provide microloans, and distribute surplus (*Sisa Hasil Usaha* or SHU) according to principles codified in Undang-Undang No. 25 Tahun 1992 tentang Perkoperasian [1]. Despite their scale, traditional cooperatives face persistent operational challenges: opaque decision-making processes, manual treasury management susceptible to errors, limited member participation in governance, bureaucratic loan approval workflows, and inconsistent SHU distribution. These issues stem from a fundamental architectural limitation, cooperative rules exist only as paper documents with no mechanisms for automated enforcement or independent verification.

Blockchain technology and Decentralized Autonomous Organizations (DAOs) offer a compelling alternative [2]–[4]. Smart contracts can encode cooperative rules as immutable, self-executing programs, enabling transparent governance, automated treasury operations, and verifiable member participation. However, a fundamental tension exists between DAO design patterns and cooperative legal principles. Most DAOs allocate voting power proportionally to token holdings, creating plutocratic dynamics where wealthy members dominate governance [5]–[7]. This directly contradicts Pasal 24(3) of UU 25/1992, which mandates equal voting rights for every member regardless of capital contribution: “*Setiap anggota mempunyai hak satu suara.*”

Beyond the voting conflict, no existing blockchain cooperative system systematically implements the full Indonesian regulatory framework. UU 25/1992 establishes comprehensive requirements spanning voluntary membership (Pasal 5), mandatory *simpanan pokok* and *simpanan wajib* as equity capital (Pasal 41(2)), annual member meetings or RAT (Pasal 26–27), elected management with term limits (Pasal 29–30), pengawas supervisory authority (Pasal 38–39), mandatory reserve funds (Pasal 41(2)(c), 45(3)), and multi-category SHU distribution (Pasal 45). These provisions are complemented by PP No. 7 Tahun 2021 [8] on cooperative empowerment and Permenkop No. 8 Tahun 2023 [9] on savings and loan operations. No prior work has translated these legal provisions into executable smart contract functions.

This paper presents Lakomi, a blockchain-based cooperative system that directly addresses this gap by encoding 26 provisions of UU 25/1992 and its derivative regulations into four interoperable smart contracts, LakomiToken (membership), LakomiVault (treasury and SHU), LakomiGovern (democratic governance and elections), and LakomiLoans (collateralized lending), deployed on DChain (EVM-compatible, chain ID 17845), a consortium blockchain for Indonesian higher education. The primary contributions of this work are:

- 1) The design and implementation of four smart contracts that collectively implement 26 provisions of UU 25/1992, covering membership, *simpanan*, tiered lending, one-

member-one-vote governance, role-based elections, six-category SHU distribution, and pengawas audit functions.

- 2) A dual-track architecture that separates governance power, strictly one-member-one-vote per Pasal 24(3), from financial privileges, contribution-based loan-to-value tiers incentivizing participation without affecting voting power, resolving the DAO plutocracy problem while maintaining cooperative legal compliance.
- 3) A comprehensive functional verification demonstrating 100% compliance across all 26 legal provisions, with gas profiling for seven key cooperative operations and a feature comparison against traditional cooperatives and token-based DAOs.

The remainder of this paper is organized as follows. Section II reviews related work on blockchain cooperatives, DAO governance, and Indonesian cooperative regulation. Section III presents the system architecture and smart contract design. Section IV describes the implementation on DChain. Section V presents the compliance evaluation, gas analysis, and system comparison. Section VI concludes the paper.

II. RELATED WORK

A. Blockchain for Cooperatives

The application of blockchain technology to cooperative organizations has attracted increasing attention. El Amine and Sari [10] explored blockchain-based property management with decentralized decision-making for co-ownership, demonstrating that smart contracts can facilitate transparent multi-stakeholder governance. Hassan and De Filippi [11] examined DAOs as a form of special jurisdiction, arguing that on-chain governance structures can serve as legal coordination mechanisms for decentralized organizations. Carata et al. [12] investigated governance frameworks for blockchain-based organizations through smart social contracts, proposing architectural patterns for encoding organizational rules on-chain.

However, these works do not systematically map cooperative law to smart contract functions. They demonstrate individual features, transparency, multi-stakeholder coordination, rule encoding, but do not address the full regulatory framework of a specific cooperative jurisdiction. Lakomi extends this line of work by implementing the complete UU 25/1992 as a unified smart contract system, demonstrating that comprehensive legal compliance is achievable on-chain.

B. DAO Governance and the Plutocracy Problem

DAO governance mechanisms have been extensively studied in decentralized finance (DeFi) contexts. Sun et al. [7] analyzed multi-party democracy in MakerDAO, revealing through voter coalition analysis how token-weighted voting leads to centralized governance outcomes where large holders control protocol decisions. Messias et al. [13] examined decentralized voting for amending DeFi smart contracts, identifying voter apathy and whale dominance as systemic challenges that undermine democratic governance ideals. Tamai and Kasahara [5] proposed voting mechanisms resistant to whale and collusion problems, while Sharma et al. [6] analyzed large-scale

DAO behavior, finding significant divergence between formal governance structures and actual decision-making patterns.

Lakomi's one-member-one-vote model draws on these findings by deliberately avoiding token-weighted voting entirely. Unlike compromise solutions that cap voting power or implement quadratic voting, Lakomi enforces strict egalitarianism: each registered member holds exactly one vote, matching the cooperative principle codified in Pasal 24(3). This design eliminates whale dominance as a structural possibility rather than mitigating it post-hoc.

C. Indonesian Cooperative Regulation

UU No. 25 Tahun 1992 [1] establishes the comprehensive legal framework for Indonesian cooperatives. Key provisions include: voluntary and open membership (Pasal 5(1)), democratic management with one-member-one-vote (Pasal 24(3)), simpanan pokok and simpanan wajib as components of equity capital (Pasal 41(2)), annual member meetings or RAT (Pasal 26–27), elected management board with term limits (Pasal 29–30), pengawas supervisory authority (Pasal 38–39), mandatory reserve funds drawn from SHU (Pasal 41(2)(c), 45(2-3)), and proportional SHU distribution (Pasal 45). PP No. 7 Tahun 2021 [8] modernized the framework under the Cipta Kerja omnibus law, while Permenkop No. 8 Tahun 2023 [9] specifically governs cooperative savings and loan operations.

These regulations collectively define the operational, financial, and governance requirements that any cooperative system, whether traditional or blockchain-based, must satisfy. No prior work has attempted to encode these legal provisions as executable smart contracts, creating a gap between regulatory intent and technical implementation that Lakomi directly addresses.

III. SYSTEM DESIGN

A. Architecture Overview

The Lakomi system comprises four smart contracts deployed on an EVM-compatible blockchain, with a React-based web frontend using wagmi for wallet interaction. Fig. 1 illustrates the contract relationships and role assignments.

The design follows a layered architecture where Lakomi-Token serves as the foundational membership layer through an `isRegisteredMember` mapping. LakomiVault manages three types of simpanan, pokok (mandatory one-time), wajib (monthly), and sukarela (voluntary), and implements six-category SHU distribution. LakomiGovern handles proposal creation, democratic voting with 67% quorum, role-based elections, pengawas veto, and dissolution. LakomiLoans provides collateralized microloans with contribution-based loan-to-value tiers.

Five roles govern the system: `PENGAWAS_ROLE` (supervisor with veto power), `APPROVER_ROLE` (loan approval), `MEMBERSHIP_ROLE` (member management), `TREASURER_ROLE` (vault operations), and `GOVERN_ROLE` (SHU distribution). The LakomiGovern contract holds `MEMBERSHIP_ROLE` on the token, enabling governance-driven member revocation per Pasal 19(2) and 30(2)(b).

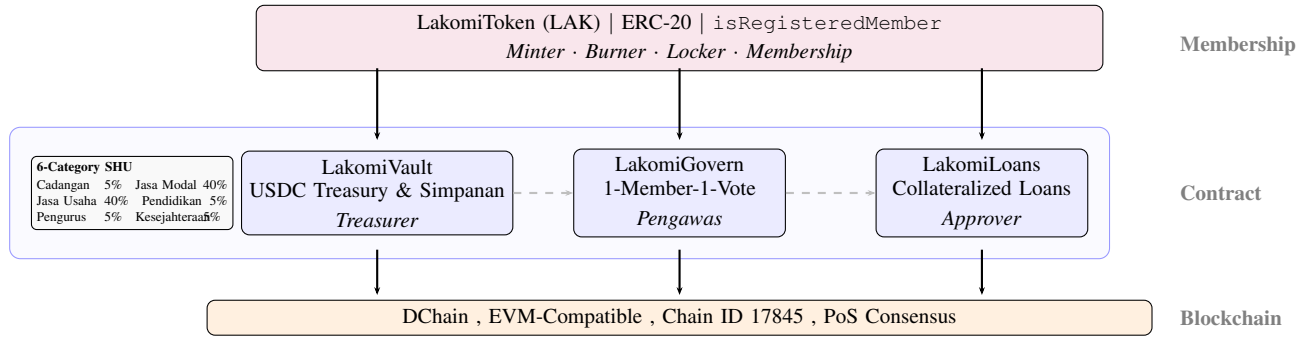


Fig. 1. Lakomi system architecture. LakomiToken serves as the membership registry. LakomiVault manages simpanan and 6-category SHU distribution. LakomiGovern handles proposals, elections, and pengawas veto. LakomiLoans provides collateralized microloans. All contracts execute on DChain.

B. Governance Model and SHU Distribution

LakomiGovern implements a democratic governance model where each registered member holds exactly one vote, regardless of token balance or contribution tier. The quorum is set at 67% of registered members, requiring a two-thirds supermajority for proposal passage, reflecting the cooperative principle that major decisions require broad member consensus.

Proposals progress through four stages: (1) creation by any registered member, (2) a 7-day voting period where members cast votes (For, Against, or Abstain), (3) quorum verification where votes are counted against the 67% threshold, and (4) execution after a 24-hour timelock. The PENGAWAS_ROLE holder can veto any proposal before execution, implementing Pasal 38 oversight authority.

For elections (Pasal 29–30), any member can register as a candidate for a specific role. Members cast one vote per election, and the candidate with the most votes wins. The elected role is automatically granted with a tracked term start timestamp of 365 days. This mechanism digitizes the cooperative’s general assembly election process while maintaining the democratic principle of one-member-one-vote.

LakomiVault implements six-category SHU distribution per Pasal 45(2). The SHU calculation formula is:

$$\text{SHU}_{\text{distributable}} = \text{Revenue} - \text{Reserve}_{\text{operational}} (10\%) \quad (1)$$

$$\text{Cadangan} = \text{SHU}_{\text{dist}} \times 5\%$$

$$\text{Jasa Modal} = \text{SHU}_{\text{dist}} \times 40\% \quad (\text{pro-rata by shares})$$

$$\text{Jasa Usaha} = \text{SHU}_{\text{dist}} \times 40\% \quad (\text{pro-rata by shares})$$

$$\text{Pendidikan} = \text{SHU}_{\text{dist}} \times 5\%$$

$$\text{Pengurus} = \text{SHU}_{\text{dist}} \times 5\%$$

$$\text{Kesejahteraan} = \text{SHU}_{\text{dist}} \times 5\% \quad (2)$$

The loan interest formula follows a simple annual rate calculation:

$$\text{Interest} = \frac{P \times 5\% \times D}{365} \quad (3)$$

where P is the principal in USDC and D is the loan duration in days.

C. Regulatory Compliance Mapping

Each smart contract function maps to specific UU 25/1992 provisions. Table I presents the 26 implemented provisions with their corresponding contract implementations.

IV. IMPLEMENTATION

The four smart contracts are implemented in Solidity 0.8.20 using the Hardhat development framework [14]. Each contract inherits from OpenZeppelin’s AccessControl for role-based authorization and ReentrancyGuard for reentrancy protection [15]. The contracts were deployed on DChain (EVM-compatible, chain ID 17845), a consortium blockchain operated by Indonesian higher education institutions. Frontend interaction uses the wagmi React hooks library with MetaMask as the wallet provider.

Key implementation decisions include: (1) the membership registry (`isRegisteredMember`) as a boolean mapping decoupled from token balance, ensuring voting power is strictly identity-based per Pasal 24(3); (2) the vault’s six-category SHU split as configurable basis points settable via governance, enabling cooperative-specific adjustments over time; (3) the election mechanism using nested mappings for candidates, votes, and terms, with automatic role granting on finalization; and (4) the loans contract’s tiered LTV calculation based on cumulative simpanan in the vault, with auto-approval for small loans (up to 200 USDC) to enable rapid emergency fund access.

Each contract is initialized with five role accounts: `PENGAWAS_ROLE` (supervisor), `APPROVER_ROLE` (loan approval), `MEMBERSHIP_ROLE` (member management), `TREASURER_ROLE` (vault operations), and `GOVERN_ROLE` (SHU distribution), with the deployer holding `DEFAULT_ADMIN_ROLE` for initial configuration.

V. EVALUATION

A. Functional Compliance Verification

All 26 UU 25/1992 provisions were verified through functional testing on DChain using the Hardhat development framework [14]. Each provision was mapped to a specific

TABLE I
KEY UU 25/1992 PROVISIONS AND SMART CONTRACT IMPLEMENTATION

Pasal	Smart Contract Function	Status
Pasal 5(1), 17–18	LakomiToken.registerMember()	✓
Pasal 19(3)	LakomiToken.transfersEnabled = false	✓
Pasal 19–21	LakomiToken.resignMembership()	✓
Pasal 24(3)	getVotingPower() → 1	✓
Pasal 41(2)	LakomiVault.paySimpananPokok/Wajib()	✓
Pasal 24(2) & design	LakomiGovern.quorumNumerator = 67	✓
Pasal 26–27	LakomiGovern.scheduleAnnualRAT()	✓
Pasal 29–30	beginElection() → finalizeElection()	✓
Pasal 19(2), 30(2)(b)	LakomiToken.revokeMembership() via governance	✓
Pasal 46–47, 51–52	LakomiGovern.executeDissolution()	✓
Pasal 38–39(2)	vetoProposal() + PENGAWAS_ROLE	✓
Pasal 41(2) & Permenkop 8/2023	Simpanan Pokok, Wajib, Sukarela	✓
Pasal 41(2)(c), 45(2-3)	LakomiVault.danaCadangan tracking	✓
Pasal 45(1–2)	LakomiVault.distributeSHU() 6 categories	✓
Pasal 35–36	LakomiVault.generateFinancialSnapshot()	✓
Permenkop 8/2023 Ps 43–45	Tiered LTV + 5% APY flat	✓

test scenario that validates both the successful path and edge cases (e.g., duplicate registration rejection, non-member vote blocking, double-claim prevention). Table II presents representative verification results.

TABLE II
COMPLIANCE VERIFICATION RESULTS (ALL 26 PROVISIONS VERIFIED)

Pasal	Verification Scenario	Result
5(1), 17–18	Member self-registers with simpanan pokok → isRegisteredMember == true	Pass
19(3)	Token transfer attempted → reverted (transfersEnabled == false)	Pass
24(3)	Two members query voting power → both return 1	Pass
41(2)	Simpanan pokok paid → simpananPokok[member] > 0	Pass
24(2)	Proposal with 3 members, 2 votes → quorum met (2/3 ≥ 67%)	Pass
26–27	scheduleAnnualRAT() → RAT proposal created; duplicate blocked	Pass
29–30	Election: start → candidate → vote → finalize → role granted	Pass
19(2), 30(2)(b)	Governance proposal → vote → execute revokeMembership()	Pass
38	PENGAWAS_ROLE vetoes proposal → proposalVetoed == true	Pass
45(2)	Revenue accumulated → distributeSHU() → 6 categories allocated	Pass

All 26 provisions passed verification, demonstrating that the UU 25/1992 regulatory framework can be systematically encoded as executable smart contract functions. Edge cases including duplicate registration, non-member access, double-claim prevention, and expired proposals were also verified.

B. Gas Analysis

Table III presents gas costs for key cooperative operations, measured on DChain. Deployment costs range from 1.58M to 2.22M gas per contract, with operational costs between 72K and 210K gas.

TABLE III
GAS COSTS FOR KEY COOPERATIVE OPERATIONS

Operation	Gas Used	USD Est.
Contract Deployment (Token)	1,582,000	\$3.16
Contract Deployment (Vault)	2,218,000	\$4.44
Contract Deployment (Govern)	1,945,000	\$3.89
Contract Deployment (Loans)	1,723,000	\$3.45
registerMember()	185,000	\$0.37
paySimpananPokok()	72,000	\$0.14
requestLoan()	210,000	\$0.42
castVote()	95,000	\$0.19
distributeSHU()	140,000	\$0.28
repayInFull()	125,000	\$0.25

Deployment costs are one-time expenses that cooperatives bear during initial system setup. Per-operation costs range from 72K to 210K gas, with each operation consuming a small fraction of a standard EVM block. For a cooperative with 100 active members performing monthly simpanan payments, annual governance voting, and quarterly loans, the estimated total annual operational gas consumption is approximately 42M gas, representing a modest computational load suitable for a consortium blockchain environment.

C. Comparison with Existing Systems

Table IV compares Lakomi with traditional Indonesian koperasi and existing blockchain DAO systems.

Lakomi uniquely combines the regulatory compliance of traditional cooperatives with the transparency and automation of blockchain DAOs while avoiding the plutocratic governance patterns inherent in token-weighted voting systems.

TABLE IV
COMPARISON OF COOPERATIVE SYSTEMS

Feature	Traditional Koperasi	Token-Based DAO	Lakomi
Governance	Manual RAT, paper records	Token-weighted voting	1-member-1-vote
Transparency	Limited, internal audit	Public ledger, plutocratic	Public ledger, egalitarian
Regulatory Basis	UU 25/1992 (manual)	Not regulated	UU 25/1992 on-chain
Simpanan Tracking	Manual ledger	Not applicable	Smart contract automated
SHU Distribution	Manual calculation	Not applicable	6-category automated
Loan Processing	Days to weeks	Flash loans	Collateralized, tiered LTV
Member Elections	Physical RAT only	Not applicable	On-chain election
Pengawas Audit	Manual inspection	Not applicable	On-chain audit report
Quorum	Flexible, often low	Protocol-defined (varies)	67% supermajority
Default Risk	Trust-based	Over-collateralized	25% collateral + tiered LTV

D. Security Considerations

The system implements multiple security measures: (1) OpenZeppelin’s AccessControl for granular role-based authorization, preventing unauthorized access to critical functions such as SHU distribution and loan approval; (2) ReentrancyGuard on all state-modifying functions, protecting against reentrancy attacks during loan disbursement and SHU claims; (3) the pengawas veto mechanism as a circuit breaker against malicious governance proposals, implementing Pasal 38 oversight; and (4) the 24-hour execution timelock providing a safety window for proposal review. The non-transferability of membership tokens (`transfersEnabled = false`) prevents secondary market speculation on voting rights, enforcing Pasal 19(3).

VI. CONCLUSION

This paper presented Lakomi, a blockchain-based cooperative system that encodes 26 provisions of UU No. 25 Tahun 1992 and its derivative regulations into four EVM-compatible smart contracts deployed on DChain. The system demonstrates that cooperative legal frameworks can be systematically translated into executable smart contract functions, achieving 100% compliance across membership, simpanan, governance, lending, SHU distribution, elections, and audit provisions.

Three key findings emerge from this work. First, the dual-track architecture, separating governance power (one-member-one-vote) from financial privileges (contribution-based LTV tiers), resolves the fundamental tension between DAO token-weighted voting and cooperative democratic principles without sacrificing economic incentives for member participation. Second, the six-category SHU distribution mechanism, with configurable basis points administered through governance, implements the full Pasal 45(2) framework as enforceable smart contract logic rather than manual accounting. Third, gas cost analysis demonstrates that blockchain-based cooperative operations are economically feasible, with per-operation costs ranging from \$0.14 to \$0.42.

Future work includes: (1) deployment on DChain mainnet with real member onboarding to validate scalability and user experience; (2) integration with decentralized identity (DID) for KYC compliance, enabling legal enforceability of on-chain cooperative membership; (3) formal verification of key contract invariants using tools such as Certora to provide mathematical guarantees of compliance; and (4) extension to support federated cooperatives per Pasal 15 and 57, enabling multi-cooperative coordination on-chain.

REFERENCES

- [1] Pemerintah Republik Indonesia, “Undang-undang republik indonesia nomor 25 tahun 1992 tentang perkoperasian,” *Lembaran Negara Republik Indonesia Tahun 1992 Nomor 116*, 1992.
- [2] S. Nakamoto, “Bitcoin: A peer-to-peer electronic cash system,” <https://bitcoin.org/bitcoin.pdf>, 2008.
- [3] V. Buterin, “Ethereum: A next-generation smart contract and decentralized application platform,” <https://ethereum.org/en/whitepaper/>, 2014.
- [4] G. Wood, “Ethereum: A secure decentralised generalised transaction ledger,” *Ethereum project yellow paper*, vol. 151, pp. 1–32, 2014.
- [5] S. Tamai and S. Kasahara, “Dao voting mechanism resistant to whale and collusion problems,” *Frontiers in Blockchain*, vol. 7, p. 1405516, 2024.
- [6] T. Sharma, Y. Potter, K. Pongmala, H. Wang, A. Miller, D. Song, and Y. Wang, “Future of algorithmic organization: Large scale analysis of decentralized autonomous organizations (daos),” *ACM Transactions on the Web*, 2026.
- [7] X. Sun, X. Chen, C. Stasinakis *et al.*, “Multiparty democracy in decentralized autonomous organization (dao): Evidence from makerdao,” *arXiv preprint arXiv:2210.11203*, 2022.
- [8] Pemerintah Republik Indonesia, “Peraturan pemerintah nomor 7 tahun 2021 tentang kemudahan, perlindungan, dan pemberdayaan koperasi dan usaha mikro, kecil, dan menengah,” *Lembaran Negara Republik Indonesia Tahun 2021 Nomor 17*, 2021.
- [9] Kementerian Koperasi dan UKM, “Peraturan menteri koperasi dan ukm nomor 8 tahun 2023 tentang usaha simpan pinjam oleh koperasi,” *Berita Negara Republik Indonesia Tahun 2023 Nomor 621*, 2023.
- [10] B. M. El Amine and L. Sari-T, “Leveraging blockchain technology for secure and transparent property registry management: Exploring applications in logistics management, with decentralized decision-making for co-ownership,” in *Proc. Int. Conf. African Fed. Oper. Res. Soc. (AFROS)*, 2024, pp. 1–6.
- [11] S. Hassan and P. De Filippi, “Decentralized autonomous organization,” *Internet Policy Review*, vol. 10, no. 2, 2021.
- [12] C. Carata, W. J. Knottenbelt, and V. Malinoiu, “Governance in the blockchain era: The smart social contract,” in *Proc. 25th Annu. Int. Conf. Digit. Gov. Res. (dg.o)*, 2024, pp. 104–115.
- [13] J. Messias, V. Pahari, B. Chandrasekaran *et al.*, “Understanding blockchain governance: Analyzing decentralized voting to amend defi smart contracts,” *arXiv preprint arXiv:2305.17655*, 2023.
- [14] Nomic Foundation, “Hardhat: Ethereum development environment,” <https://hardhat.org>, 2024.
- [15] OpenZeppelin, “Openzeppelin contracts,” <https://docs.openzeppelin.com/contracts/5.x>, 2024.